

Here's a **proper 10-mark, well-structured answer** you can directly write in your exam 📌

✍️ Tokens, Lexemes, and Patterns in Lexical Analysis

◆ Introduction

👉 **Lexical Analysis** is the first phase of a compiler. It reads the source code and converts it into a sequence of tokens.

👉 The main components involved in this process are **tokens, lexemes, and patterns**.

◆ 1. Token

👉 A **token** is a **classification or category of lexemes**.

👉 It represents a group of strings with similar meaning.

Examples of Tokens:

- **KEYWORD** → `if, while, int`
- **IDENTIFIER** → variable names like `x, sum`
- **NUMBER** → `10, 25`
- **OPERATOR** → `+, -, =`
- **SEPARATOR** → `;, ,`

👉 Tokens are usually represented as:

<Token_name, Attribute_value>

Example:

<IDENTIFIER, x>

<NUMBER, 10>

◆ 2. Lexeme

👉 A **lexeme** is the **actual sequence of characters** in the source code that matches a token pattern.

👉 It is the **smallest unit of a program**.

Example:

For the statement:

```
int x = 10;
```

Lexemes are:

int, x, =, 10, ;

👉 Each lexeme corresponds to a token.

◆ 3. Pattern

👉 A **pattern** is a **rule or regular expression** that describes the structure of lexemes for a token.

👉 Patterns are used by the lexical analyzer to identify tokens.

Examples:

- IDENTIFIER $\rightarrow [a-zA-Z][a-zA-Z0-9]^*$
 - NUMBER $\rightarrow [0-9]^+$
 - KEYWORD $\rightarrow$ fixed strings like `if`, `while`
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◆ Combined Example

For the statement:

```
int count = 20;
```

Lexeme	Token	Pattern
int	KEYWORD	fixed word
count	IDENTIFIER	<code>[a-zA-Z][a-zA-Z0-9]*</code>
=	OPERATOR	<code>=</code>
20	NUMBER	<code>[0-9]+</code>
;	SEPARATOR	<code>;</code>

◆ Relationship

👉 The relationship between them is:

- **Pattern** defines rules
 - **Lexeme** matches the pattern
 - **Token** represents the category
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◆ Conclusion

👉 Tokens, lexemes, and patterns are fundamental concepts in lexical analysis that help the compiler break the source code into meaningful units for further processing.

Final One-Line Answer

👉 A token is a category of lexical units, a lexeme is the actual character sequence, and a pattern is a rule that defines the structure of lexemes.

If you want, I can give you a **handwritten-style answer or diagram for extra marks** 👍